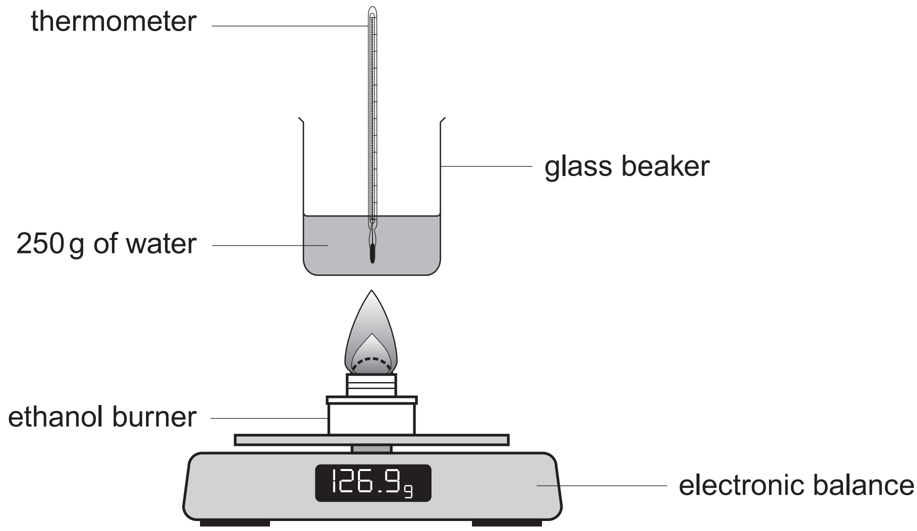
8. Mountaineers often choose an ethanol burner when hiking in extremely cold conditions.

A group of students was asked to investigate how the mass of ethanol burned is related to the amount of energy given out.



Four students each burned a different mass of ethanol and recorded the temperature rise of 250 g of water. The diagram shows the apparatus used by each of the students.
They used their results to calculate the energy given out.

The table shows the students' results.

Student	Mass of ethanol burned (g)	Energy given out ($J \times 10^4$)
A	1.1	2.2
B	1.8	3.6
C	2.9	5.8
D	4.2	8.4

Examiner only

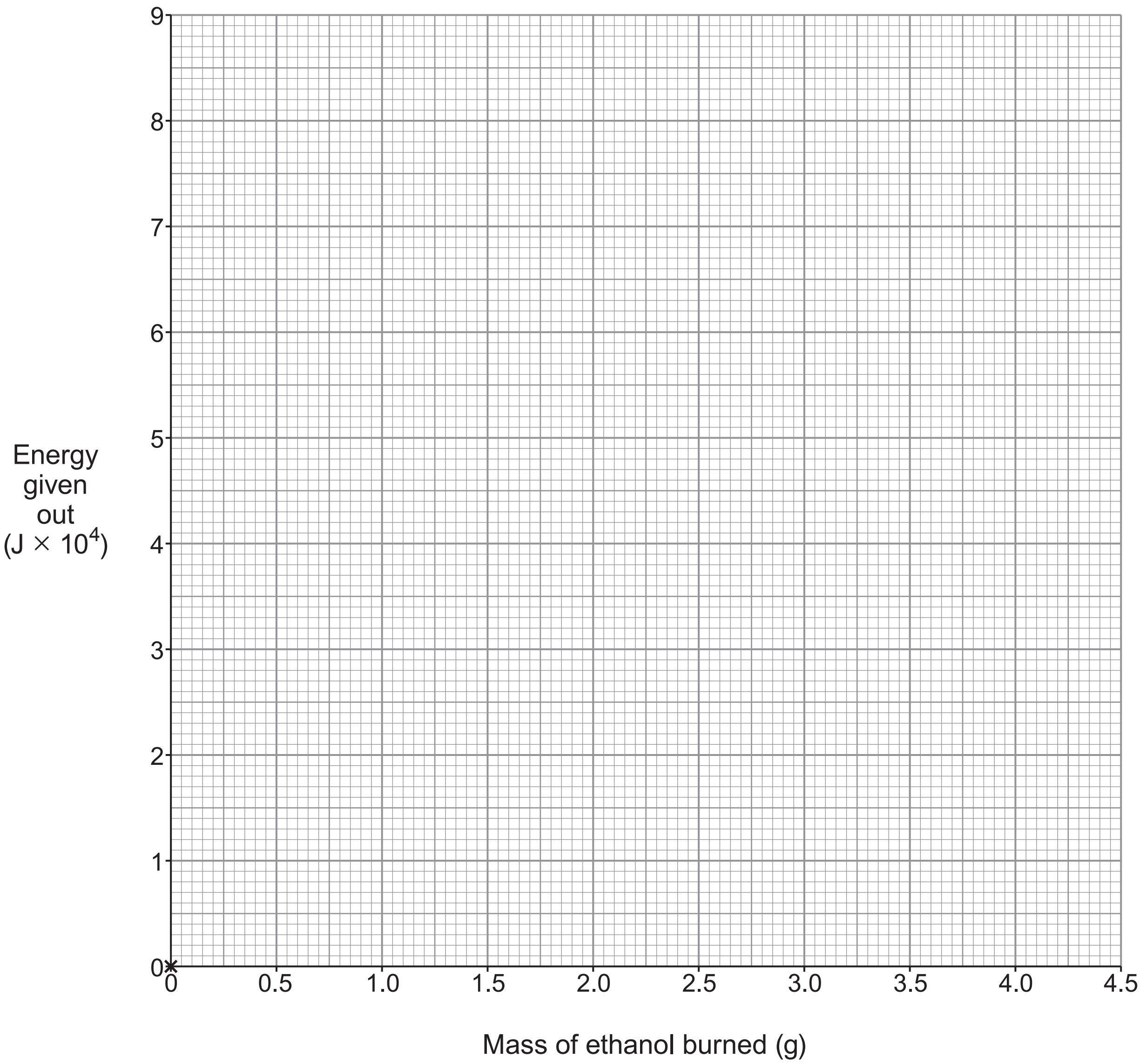
(a) The temperature rise can be calculated using the formula below.

energy given out (J) = 4.2 × temperature rise (°C) × mass of water (g)

Use this formula to calculate the temperature rise recorded by student **D**. [2]

Temperature rise = ... °C

(b) Plot the energy given out against mass of ethanol burned on the grid below and draw a suitable line. [3]



Examiner only

(c) Describe the relationship between the mass of ethanol burned and the energy given out. [2]

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(d) All the recorded temperature rises were **lower** than expected.

Explain **one** piece of advice you would give the students to resolve this problem. [2]

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